TRANSFERABILITY OF FAIRNESS-AWARE MACHINE LEARNING: AN EMPIRICAL EVALUATION OF ADAFAIR ON SMALL-SCALE IMBALANCED DATA

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THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE AND
ARTIFICIAL INTELLIGENCE
AT THE SCHOOL OF HUMANITIES AND DIGITAL
SCIENCES OF TILBURG UNIVERSITY

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DATE

June 23rd, 2025

WORD COUNT

7964

ACKNOWLEDGMENTS

I want to thank my thesis supervisor, Dr Travis J Wiltshire, for being very helpful during our meetings and guiding me through this thesis. And my girlfriend for her constant support and for helping me tackle all the formatting challenges.

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Abstract

This study investigates the transferability of the AdaFair algorithm, a fairness-aware boosting classifier, to small datasets, focusing on its ability to maintain balanced accuracy and fairness under class imbalance. Unlike prior research, which primarily tested AdaFair on large datasets, this work evaluates its performance on the large KDD census income dataset (138,335 instances) and the smaller Communities and Crime dataset (1,994 instances), both highly imbalanced. The methodology involves comparing AdaFair against AdaBoost, applying upsampling/ downsampling, and examining the dataset structure to explore subgroup distributions. Findings reveal that AdaFair excels on the KDD census income dataset but struggles on the Communities and Crime dataset, failing to classify positive non-protected instances due to extreme proportion and distance imbalances. Upsampling strategies did not mitigate fairness issues, and downsampling showed a dataset size threshold below which fairness deteriorates. These results reveal AdaFair's limitations when applied to small, imbalanced datasets and highlight distance imbalance as a key contributing factor. This insight is crucial for developing fairness-aware algorithms that can mitigate bias in real-world applications, particularly in domains like biomedicine where data is limited.

DATA SOURCE, ETHICS, CODE AND TECHNOLOGY STATEMENT

Data Source: The KDD census income dataset has been acquired from the UCI Machine Learning Repository (2000). The Communities and Crime dataset has also been acquired from UCI Machine Learning Repository (Redmond 2002). The obtained data is anonymised. Work on this thesis did not involve collecting data from human participants or animals. The original owner of the data and code used in this thesis retains ownership of the data and code during and after the completion of this thesis. All the figures belong to the author. The thesis code can be accessed through the GitHub repository via the following reference (Carree, 2025). Part of the CODE has been adapted by the author from (Iosifidis & Roy, 2023), licensed under a CC0 BY 4.0. The reused/ adapted code fragments are clearly indicated in the notebook. In terms of writing, the author used assistance with the language of the paper. A generative language model, Chat-GPT version 4o, (OpenAI, 2023) was used to improve the author's original content, for paraphrasing, spell checking, and grammar. Moreover, Grammarly was used as a spell and grammar checker within Word (Grammarly Inc., n.d.). The citations were organized using Citavi version 7.0.7.

1 Introduction

Businesses, governments, and other organizations widely employ algorithms that utilize artificial intelligence (AI) to make decisions that significantly impact our everyday lives. Moreover, the rise of large language models has made AI an increasingly common part of our daily lives, particularly through AI personal assistants such as chatbots and voice-activated tools like Siri and Alexa (Maynard, 2025). These technologies have the potential to enhance our lives by offering solutions to significant societal challenges, such as transforming the healthcare industry by improving diagnosis and personalizing treatment while reducing costs (European Commission, 2025b). However, the reliance on these algorithms also entails the risk of getting rejected for a job or medical treatment (Ntoutsi et al., 2020). This can lead to discriminatory behaviour towards certain population groups, which has already been observed in multiple cases. For example, the COMPAS tool, used to assess the likelihood of recidivism, was found to assign higher risk scores to Black defendants (racial bias) (Larson et al., 2016). Similarly, in Amazon's Prime case, an AI algorithm was tasked with determining which areas of a city were eligible for advanced services, and neighborhoods predominantly inhabited by Black residents were overlooked. The model demonstrated racial bias, even though race was not explicitly included as a feature (Ingold & Soper, 2016). Moreover, Google's ad tool showed significantly fewer high-paying jobs for women compared to men (gender bias) (Datta et al., 2015). Such incidents have led to an increase in public concern, with 52% of Americans being more concerned than excited about the increasing use of AI (Tyson & Kikuchi, 2023). From public administration, there has also been increased attention to these problems, with the European Union launching the AI Act in 2024 to promote trustworthy AI (European Commission, 2025a). Among the seven ethical principles established to promote trustworthy AI, the principles of diversity, non-discrimination, and fairness are fundamental (European Union, 2024).

Despite the emerging field of bias mitigation and fair algorithms offering numerous opportunities, it remains in its infancy, presenting several unresolved challenges. A critical yet underexplored issue is the transferability of fairness-aware algorithms across datasets, particularly smaller ones, which are prevalent in domains like biomedicine, where data collection is constrained by the need for expert annotation (Wang et al., 2023). Additional challenges include the lack of consensus on fairness definitions, with over 109 metrics in use and no standardized method for selection (Hort et al., 2024). As well as the scarcity of datasets, unreliable or inaccessible protected attributes, and shifting data distributions in real-world applications. Moreover, most research remains limited to binary classification, with few protected attributes, leaving multi-class problems and nuanced demographic factors largely unaddressed (Hort et al., 2024). This paper focuses on the transferability challenge, specifically investigating the AdaFair algorithm (Iosifidis & Ntoutsi, 2019), a fairness-aware boosting method built on AdaBoost. AdaFair addresses class imbalance by adjusting instance weights per boosting round using a cumulative fairness measure based on Equalized Odds, aiming to balance false positive and false negative rates across protected and non-protected groups. While it has shown promise on large datasets such as Adult census income (Becker & Kohavi, 1996), Bank marketing (Moro et al., 2014),

COMPAS (Larson et al., 2016), and KDD census income (UCI Machine Learning Repository, 2000), its performance on small datasets remains largely untested. This gap is significant, as large, annotated datasets are scarce, and real-world applications often rely on smaller, imbalanced datasets where fairness and accuracy must still be ensured (Wang et al., 2023). To address this, the thesis examines how dataset size impacts AdaFair's performance, with a particular emphasis on small, imbalanced datasets. The research question is: *How is the performance of the AdaFair algorithm influenced by dataset size, particularly for small datasets, in terms of balanced accuracy and equalized odds?* This is explored by comparing AdaFair's performance on the large KDD census income dataset (138,335 instances) and the smaller Communities and Crime dataset (1,994 instances), both highly imbalanced. In the KDD census income dataset, the protected group is defined as females, while in the Communities and Crime dataset, it consists of communities with a Black population exceeding 6%. To better understand the underlying factors influencing the results, AdaFair is applied to upsampled versions of the Communities and Crime dataset and downsampled versions of the KDD census income dataset. Additionally, a Principal Component Analysis (PCA) (Pearson, 1901; Jolliffe & Cadima, 2016) was conducted on the Communities and Crime dataset to examine its underlying data structure and investigate whether distributional patterns, specifically, particular types of imbalances, could help explain the observed performance issues.

2 Related work

This section provides an overview of key concepts and methods used in the creation of fairness-aware machine learning. It starts by outlining an expanded taxonomy of class imbalance, emphasizing how different forms of imbalance can impact both model fairness and performance. The discussion then turns to fairness metrics, with particular attention to Equalized Odds, the primary metric adopted in this study. Following that, the section reviews common bias mitigation strategies, categorized by their application stage: pre-processing, in-processing, and post-processing, and illustrates each with representative approaches. Finally, it introduces the AdaFair algorithm, detailing its fairness-driven enhancements to AdaBoost and its mechanisms for handling class imbalance, laying the groundwork for the experiments presented in this work.

2.1 Class imbalance

In machine learning, class imbalance is a common challenge that can compromise fairness and hinder generalizability. Typically, this is viewed as a problem of imbalance in the proportions of classes (Wu, 2023). However, a new taxonomy was proposed by Wu (2023), which expands this concept to include five types of imbalances: proportion, variance, distance, neighborhood, and quality. Furthermore, a distinction is made by Wu (2023) between global and local imbalances. Below, each type is explained in detail.

Proportion imbalance

Proportion imbalance is the most familiar type, occurring when classes have unequal numbers of samples (Wu, 2023). For instance, in email spam detection, legitimate emails typically comprise 95% of the dataset while spam emails represent only 5%. This disparity can lead models to develop a bias toward the

majority class, achieving seemingly high accuracy by predominantly predicting the majority class while failing to effectively identify minority class instances (Wu, 2023).

Variance Imbalance

Variance imbalance occurs when different classes exhibit significantly different levels of data variability (Wu, 2023). For instance, in a music genre classification task, classical music recordings might have low variance due to their structured composition. While jazz recordings might display high variance because of improvisation. This imbalance creates challenges for machine learning models, which may become overconfident when predicting the low-variance classical pieces and underconfident or inaccurate when classifying the highly variable jazz tracks. The problem intensifies when the minority class (such as jazz) exhibits higher variance, as the model receives limited exposure to the full spectrum of possible musical expressions, ultimately compromising its ability to generalize effectively (Wu, 2023).

Distance Imbalance

Distance imbalance refers to varying separability between classes in the feature space, where some class pairs are naturally closer together than others (Wu, 2023). In handwritten digit recognition, distinguishing between '6' and '9' proves challenging due to their similar curved structures, while separating '1' and '0' is relatively straightforward given their distinct geometric features. Models typically excel at classifying well-separated class pairs but struggle with those having overlapping feature representations (Wu, 2023).

Neighborhood Imbalance

Neighborhood imbalance is critical in graph-based or relational data tasks, where a data point's classification depends on its neighbors (Wu, 2023). In a social network, a user's class might be influenced by their connections. If a user is surrounded by neighbors from a different class, misclassification becomes more likely. This imbalance is prominent in node classification with graph neural networks, requiring methods that account for local structure to ensure fair predictions and mitigate the influence of imbalanced neighborhoods (Wu, 2023).

Quality Imbalance

Quality imbalance emerges when different classes exhibit varying levels of data reliability, stemming from inconsistent noise levels, incomplete information, or annotation mistakes (Wu, 2023). For example, in a medical dataset, certain conditions suffer from higher rates of incorrect labels because they are more difficult to diagnose, which distorts how the model learns to recognize certain conditions. This issue is frequently underestimated yet remains crucial, particularly in datasets compiled from multiple sources where certain classes may have undergone more rigorous data collection than others (Wu, 2023).

Global vs. Local Imbalances

The taxonomy by Wu (2023) also distinguishes between global and local imbalances. Global imbalances affect the entire dataset, such as overall class proportions. Local imbalances occur within specific subsets, like underrepresented subgroups within a class. Local imbalances are subtler and require more attention to mitigate (Wu, 2023).

The AdaFair algorithm uses the definition of class imbalance as the ratio between positive and negative classes (Iosifidis & Ntoutsi, 2019), referring to the proportion imbalance in the taxonomy of Wu (2023). The importance of this will be clear when discussing the results of this study.

2.2 Fairness metrics

Fairness metrics play a crucial role in the bias mitigation process (Hort et al., 2024), as they quantify the level of bias both before and after applying mitigation techniques, thereby helping assess the effectiveness of those methods. As previously noted, there are at least 109 different fairness metrics, reflecting the complexity and nuance of this research area. These metrics are typically grouped into six categories (Hort et al., 2024), each based on a different perspective: dataset labels, predicted outcomes, predicted vs. actual outcomes, predicted probabilities, similarity, and causal reasoning. Label-based metrics evaluate bias within the dataset itself by examining whether protected and unprotected groups are fairly represented (Hort et al., 2024). Predicted outcome-based metrics, such as *Demographic Parity* and *Disparate Impact*, assess whether different groups receive a similar proportion of positive outcomes (Hort et al., 2024). Predicted vs. actual outcome-based metrics measure performance disparities between groups, especially error rates, with *Equalized Odds* and *Equality of Opportunity* being the most widely used (Hort et al., 2024). Probability-based metrics, like *Strong Demographic Disparity*, consider differences in predicted probabilities across groups (Hort et al., 2024). Similarity-based metrics focus on ensuring that individuals with similar input features receive similar predictions, often using distance measures like k-nearest neighbors (Hort et al., 2024). Finally, causal reasoning-based metrics employ causal graphs to detect direct or indirect influence of sensitive attributes on outcomes (Hort et al., 2024). In this study, the AdaFair algorithm uses *Equalized Odds* as its fairness metric, due to its demonstrated effectiveness in recent literature (Iosifidis & Ntoutsi, 2019). *Equalized Odds* measures the prediction error disparity between protected and non-protected groups. The following notation is used to distinguish between protected and non-protected groups within the positive and negative classes s+, s− for the protected group, and $\overline{s^{-}}$+, $\overline{s^{-}}$− for the non-protected group (Iosifidis & Ntoutsi, 2019). The differences between **false positive rates (FPR)** and **false negative rates (FNR)** across these groups are calculated as follows, where $\hat{y}$ denotes the predicted label:

$$\delta_{\mathrm{FPR}} = P(y \neq \hat{y} \mid \overline{s^{-}}) - P(y \neq \hat{y} \mid s^{-})$$

$$\delta_{\mathrm{FNR}} = P\left(y \neq \hat{y} \mid \overline{s^{+}}\right) - P(y \neq \hat{y} \mid s^{+})$$

The goal is to minimize both differences:

$$\text{Equalized Odds} = |\delta_{\mathrm{FPR}}| + |\delta_{\mathrm{FNR}}|$$

The value range for each of $|\delta\ FPR|$, $|\delta\ FNR|$ is [0,1], where 0 stands for no discrimination and 1 stands for maximum discrimination. Equalized Odds values lie in the [0-2] range (Iosifidis & Ntoutsi, 2019).

2.3 Bias mitigation methods

In the literature, three different types of bias mitigation methods are distinguished based on the stage of the machine learning process they are applied in: pre-processing, in-processing, and post-processing (Hort et al., 2024). However, these are not mutually exclusive. For example, an approach might involve applying pre-processing techniques before modifying the training procedure during the in-processing stage. In addition, multiple methods at the same stage of the process could be used, like two preprocessing methods could be utilized simultaneously (Hort et al., 2024).

2.3.1 Pre-processing

One of the most common sources of discrimination in machine learning stems from historical bias embedded in training data (Hort et al., 2024). Pre-processing methods aim to address this issue by changing the data before training. These techniques are typically model agnostic, meaning they can be combined with any classifier after the data has been adjusted (Hort et al., 2024). These can be divided into 4 groups: relabeling and perturbation, sampling, latent variables, and representation. Relabeling and perturbation methods apply changes to the values of the training data (Hort et al., 2024). With relabeling, modifying ground truth labels, while perturbation alters the remaining features in the training data. Massaging (Kamiran & Calders, 2009), a popular relabeling technique, employs a ranker to identify optimal relabeling candidates. Specifically, it targets instances near the decision boundary to reduce accuracy loss from relabeling. Sampling approaches rebalance the data by changing the distribution of samples or adapting their impact in training (Hort et al., 2024). The rebalancing of the data can be achieved in multiple ways: removal of datapoints from the dominant class (downsampling) or addition of datapoints from the minority class (upsampling). For upsampling, instances of the minority group can be duplicated (Hort et al., 2024), or Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002) can be applied, which generates similar instances based on existing ones. Latent variables augment the data with extra, preferably unbiased features (Hort et al., 2024). For example, Calders and Verwer (2010) used clustering with an expectation maximization algorithm to designate positive or negative latent labels, whereas Gupta et al. (2018) tackled absent group labels by creating a latent variable for group membership and embedding fairness constraints in the training process. Lastly, representation learning transforms the data to reduce bias while keeping as much information as possible (Hort et al., 2024). The first of this type of approach was Learning Fair Representations (LFR) (Zemel et al., 2013), which formulates representation learning as an optimization problem with two main objectives: removing information related to protected attributes and preserving as much information as possible from non-sensitive features. Since then, a multitude of approaches have been proposed (Hort et al., 2024). For this study, downsampling and both types of upsampling approaches will be utilized in the exploration of the results.

2.3.2 In processing

In-processing methods intervene during the model training phase, incorporating fairness directly into the learning process (Hort et al., 2024). These can also be divided into 4 groups: regularization and constraints,

adversarial learning, compositional models, and adjusted learning. Both regularization and constraints apply a change to the algorithm's loss function (Hort et al., 2024). Regularization does this by adding a term to the loss function that penalizes discrimination. Constraints determine a specific bias level, which cannot be violated during training. Most of these methods use group fairness decisions; however, they have also been applied to obtain individual fairness. As well as to achieve fairness in cases with multiple protected attributes or adding to existing regularization techniques like the L2 norm (Hort et al., 2024). Adversarial learning trains two models simultaneously, with one trained to obtain good classification performance while the adversary is trained to take fairness into account (Hort et al., 2024). Then both models compete against each other to improve performance. One of these is Adversarially Reweighted Learning (ARL) (Lahoti et al., 2020), where the adversary adjusts instance weights based on regions in the input space with high training loss. Thereby promoting difficult or biased regions to improve fairness and model robustness. Compositional approaches utilize multiple classification models together (Hort et al., 2024); some use a specific model per group (protected, non-protected) while others use the ensemble architecture. Although training separate classification models for protected and non-protected groups can improve accuracy within each group, at the same time, it reduces the amount of training data (Hort et al., 2024). To mitigate this issue, transfer learning (Dwork et al., 2016) was employed by training classifiers on data from the target group, supplemented with data from other groups assigned reduced weights. Finally, adjusted learning methods try to mitigate bias by changing existing algorithms or creating novel ones (Hort et al., 2024). This has been applied to a variety of classification models, such as Bayesian models, Markov Random Fields, Neural Networks, Decision Trees, bandits, boosting, and Logistic Regression (Hort et al., 2024). The AdaFair algorithm falls under this last category as it adjusts the Adaboost algorithm.

2.3.3 Postprocessing

Although not the primary focus of this study, post-processing remains relevant while it offers strategies to improve fairness after model training (Hort et al., 2024). These techniques can be particularly valuable when pre- or in-processing methods like AdaFair fall short. Post-processing methods can be divided into two categories: **white-box** and **black-box** approaches (Hort et al., 2024). White-box methods modify the internal behavior of the classifier, typically by adjusting its decision boundary. For example, Fish et al. (2016) proposed shifting the decision boundary of AdaBoost to reduce discrimination, while Pedreschi et al. (2009) adjusted the confidence of classification rules in the CPAR algorithm to improve fairness. In contrast, black-box methods operate solely on the classifier's outputs, without altering the model itself (Hort et al., 2024). A prominent example is the method by Hardt et al. (2016), which sets prediction thresholds to equalize error rates between protected and non-protected groups. An extension of this work, Pleiss et al. (2017) introduced a calibration technique to maintain equal error rates across groups.

2.4 The AdaFair algorithm

The AdaFair algorithm, proposed by Iosifidis and Ntoutsi (2019), is a fairness-aware boosting classifier designed to address both class imbalance and

discrimination in machine learning. Built upon the AdaBoost framework, AdaFair introduces modifications to promote fairness, specifically targeting Equalized Odds, a fairness metric that minimizes disparities in false positive and false negative rates between protected and non-protected groups (Iosifidis & Ntoutsi, 2019), as mentioned before. Unlike traditional AdaBoost, which focuses solely on minimizing classification error (Schapire, 1999), AdaFair incorporates a cumulative fairness notion, adjusting instance weights in each boosting round based on the fairness behavior of the ensemble up to that point (Iosifidis & Ntoutsi, 2019). This cumulative approach ensures that fairness is evaluated across all weak learners, promoting parity in true positive rates (TPR) and true negative rates (TNR) for both groups.

AdaFair also addresses class imbalance by optimizing for the balanced error rate (BER) rather than the standard error rate (ER), which tends to favor the majority class in imbalanced datasets (Iosifidis & Ntoutsi, 2019). With the parameter c controlling the trade-off between BER and ER, and the model optimizing solely for BER when c is set to 1. The algorithm employs confidence scores in its reweighting process, prioritizing instances misclassified with higher confidence to enhance decision boundary robustness (Iosifidis & Ntoutsi, 2019). The final ensemble is optimized by selecting the sequence of weak learners that balances predictive performance and fairness, using 200 boosting rounds with Decision Stumps as weak learners. The AdaFair algorithm operates under the following assumptions (Iosifidis & Ntoutsi, 2019): a dataset D consisting of n independently and identically distributed (i.i.d.) samples drawn from a joint distribution $P(F, S, y)$, where F denotes **non-sensitive attributes,** S represents **sensitive attributes** (e.g., gender or race), and y is the class label. A binary classification task is assumed with $y \in \{+, -\}$, and a single binary sensitive attribute $S \in \{s, \bar{s}\}$, where s and $\bar{s}$ correspond to the **protected** and **non-protected** groups (e.g., female, male), respectively. The notation s+, s− and $\bar{s}$+, $\bar{s}$− is used to refer to members of the protected and non-protected groups with positive (negative) class labels, respectively. The goal of the algorithm is to learn a classification function f : (F, S) → y that accurately predicts class labels for unseen data instances. For a more detailed explanation of the algorithm's underlying mechanism, readers are encouraged to consult the original paper by Iosifidis and Ntoutsi (2019).

Experimental results on datasets like KDD census income (UCI Machine Learning Repository, 2000) and COMPAS (Larson et al., 2016) demonstrate AdaFair's superiority over fairness-agnostic methods (e.g., vanilla AdaBoost (Schapire, 1999), SMOTEBoost (Chawla et al., 2003)) and other fairness-aware approaches (e.g., Krasanakis et al., 2018; Zafar et al., 2017), achieving up to 25% improvement in balanced accuracy while maintaining low Equalized Odds (Iosifidis & Ntoutsi, 2019). However, its performance on small, highly imbalanced datasets remains underexplored, motivating this study's focus on transferability to these datasets.

3 Method

In this chapter, the methods used to evaluate the AdaFair algorithm against the standard AdaBoost algorithm for classification tasks on two datasets, Communities and Crime, and KDD census income, will be explained. The primary objective is to assess AdaFair's ability to balance predictive accuracy with fairness, particularly concerning protected attributes (percentage of black population in Communities and Crime, sex in KDD

census income) in situations of data scarcity. As required by AdaFair, the class labels were encoded as 1 for positive instances and -1 for negative instances (Iosifidis & Roy, 2023). Each dataset was randomly divided into training and test sets using a 50/50 split, consistent with the original paper (Iosifidis & Ntoutsi, 2019). The reported results are averaged over five independent runs, which is half the number used in the original paper (Iosifidis & Ntoutsi, 2019), to reduce computational time. To ensure a fair comparison, the same parameters as in the original paper (Iosifidis & Ntoutsi, 2019) were used, and no hyperparameter tuning was performed. Notably, AdaBoost was applied exclusively to the original datasets and was not evaluated on the upsampled or downsampled variants. To facilitate a visual analysis of model performance, the results were presented using plots. PCA was employed to investigate the underlying factors contributing to AdaFair's performance on the Communities and Crime dataset. PCA was selected to visualize the distribution of the four subgroups (positive protected, negative protected, positive non-protected, negative non-protected) in a reduced feature space, enabling the identification of potential distributional imbalances beyond proportional differences. The 20-dimensional feature space, excluding both the protected attribute and the class label, was projected onto the first two principal components, which capture the greatest variance in the data (Pearson, 1901; Jolliffe & Cadima, 2016). This projection enabled the examination of whether the positive, non-protected group exhibited distinct patterns that might explain AdaFair's difficulty in correctly classifying these instances. The analysis included calculating group centroids, variances along each principal component, and 95% confidence ellipses based on the covariance structure of each subgroup. This visualization approach enables the assessment of multiple types of imbalances from Wu's taxonomy (2023), including variance imbalance and distance imbalance, which are critical for going beyond simple proportional imbalance in the search for explanations of the observed performance.

3.1 Datasets

KDD census income dataset

The KDD census income dataset, obtained from the UCI Machine Learning Repository (2000), is derived from 1994 U.S. Census data, capturing demographic and economic information. After combining the training and test sets, the dataset contained 299,285 instances. Following preprocessing, this was reduced to 138,335. It includes 42 features, such as age, education, sex (protected attribute), and income level (renamed class, the target variable). The target variable was binarized: instances with income over $50,000 were labeled as 1, and others as -1. The protected attribute 'sex' was binarized as 0 for males and 1 for females, with females designated as the protected group. Descriptive statistics reveal 72125 protected and 66210 non-protected instances, with 130221 positive and 8114 negative class labels, indicating significant class imbalance. This dataset was selected due to its use in the AdaFair paper and its large class imbalance.

Communities and Crime dataset

The Communities and Crime dataset, also sourced from the UCI Machine Learning Repository (Redmond, 2002), contains aggregated socio-

economic and crime statistics from 1990s U.S. communities. After preprocessing, the dataset comprises 1994 records. Twenty-two relevant features were selected, including racepctblack (protected attribute), pctWInvInc (percentage with investment income), PctUnemployed, and class (target variable). The target variable class was binarized as 1 (*ViolentCrimesPerPop* ≥ 0.7), indicating high crime rates, and -1 otherwise, indicating low crime rates. To conform with AdaFair's binary input requirements, the sensitive attribute racepctblack was binarized such that communities with a Black population of 6% or more were assigned to the protected group (encoded as 1), and those with less than 6% were assigned to the non-protected group (encoded as 0). This resulted in 1,038 protected and 956 non-protected instances. The class distribution includes 122 positive and 1,872 negative labels, reflecting a class imbalance similar to that of the KDD census income dataset. This dataset was selected to compare AdaFair's behavior under similar class imbalance conditions but with significantly fewer instances, allowing testing of the models' performance in low-data environments.

For a more in-depth analysis of both datasets, the paper by Le Quy et al. (2022) on the most common datasets in fairness-aware machine learning can be consulted.

3.2 Data preprocessing

KDD census income dataset

For the Census Income dataset, preprocessing concatenated the training and test sets to enable custom train-test splits. The protected attribute sex was mapped to 0 (Male) or 1 (Female), and the target variable income_level was binarized as -1 (≤ $50,000) or 1 (> $50,000). Categorical features (e.g., education, marital_status) were label-encoded to numerical values, as the models did not accept categorical variables. Missing values were dropped, and duplicates were removed.

Communities and Crime dataset

Preprocessing for the Communities and Crime dataset involved several steps to ensure compatibility with AdaFair. Following the approach outlined by Heidari et al. (2018), 22 of the original 127 attributes were retained after excluding all columns with missing values. The target variable, ViolentCrimesPerPop, was binarized using a threshold of 0.7, and the protected attribute racepctblack was binarized at 0.06 (representing a 6% Black population), following the methodology used in Kearns et al. (2018). Duplicate rows were removed to prevent overfitting. For both datasets, each instance was assigned to one of four groups: positive_protected, negative_protected, positive_non_protected, and negative_non_protected, indicating whether they belong to the positive or negative class and the protected or non-protected group.

3.3 KDD census income downsampling

Two downsampled versions of the KDD census income dataset were created to assess the impact of dataset size on performance, containing 10- and 100-times fewer instances than the original. This was done using the train_test_split method from scikit-learn. The stratify parameter in train_test_split was utilized to preserve the original distribution of class

labels and protected group membership during downsampling. This ensured that the class and group proportions remained consistent across the downsampled datasets, allowing for a fair comparison of model performance and fairness across different dataset sizes.

3.4 Communities and Crime upsampling

To address the class imbalance in the Communities and Crime dataset and the severe underrepresentation of the positive non-protected group, three distinct upsampling strategies were employed. The first approach involved upsampling the entire dataset by duplicating each instance according to fixed upsampling factors of 2, 10, and 100. This method aimed to assess the effect of general data enlargement on model performance and fairness. The second approach specifically targeted the positive non-protected group, duplicating only these instances using the same upsampling factors. The third strategy employed SMOTE (Chawla et al., 2002) to generate synthetic samples for the positive non-protected group by interpolating between existing instances based on their k-nearest neighbors. Due to the limited number of positive non-protected instances, k_neighbors was set to 1. This approach aimed to balance the group sizes by matching the number of positive non-protected samples to that of the positive protected group. These strategies were designed to evaluate whether upsampling could mitigate the fairness limitations observed in the Communities and Crime dataset.

3.5 Model specifications

AdaBoost

The standard AdaBoost algorithm (Schapire, 1999), implemented via scikit-learn's AdaBoostClassifier, serves as a baseline to highlight AdaFair's fairness improvements. It uses a DecisionTreeClassifier with max_depth=1, with the number of estimators matched to AdaFair for fair comparison. AdaBoost was selected due to its use as a baseline during the introduction of the AdaFair algorithm (Iosifidis & Ntoutsi, 2019).

AdaFair

AdaFair extends AdaBoost by incorporating fairness constraints during training. It uses a DecisionTreeClassifier as the base estimator with default parameters except max_depth=1. The number of estimators was set to 200, and the trade_off_c parameter was set to 1, considering only the balanced error rate, following the original paper (Iosifidis & Ntoutsi, 2019).

3.6 Implementation details

Both models were implemented using Python 3.8.10 (Python Software Foundation, 2019) to ensure compatibility with AdaFair's dependencies (Iosifidis & Roy, 2023). The implementation relied on several key libraries: pandas (McKinney, 2010; The pandas development team, 2020) for data manipulation, scikit-learn (Pedregosa et al., 2011) for AdaBoost, preprocessing, and evaluation metrics, imbalanced-learn (Lemaître et al., 2017) for the SMOTE algorithm, and seaborn (Waskom, 2021) and Matplotlib (Hunter, 2007) for data visualization. The custom AdaFair library (Iosifidis & Roy, 2023) was also integrated. To ensure the reproducibility of results, a random seed of 42 was applied to all

randomized processes, including dataset splitting and SMOTE. For model evaluation, StratifiedShuffleSplit was used to generate five train-test splits, each with a 50:50 ratio.

3.7 Evaluation

The model's performance was evaluated using a combination of accuracy and fairness metrics, each of them being utilized by the AdaFair paper (Iosifidis & Ntoutsi, 2019). All of them will be explained below:

- Accuracy: This metric represents the proportion of correctly classified instances across all data, calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN denote the total true positives, true negatives, false positives, and false negatives, respectively.

- Balanced Accuracy: To address potential class imbalance, balanced accuracy was computed as the average of the true positive rate (TPR) and true negative rate (TNR) across all instances:

$$\text{Balanced Accuracy} = \frac{\text{TPR} + \text{TNR}}{2}$$

Here, TPR measures the proportion of positive instances correctly identified:

$$\text{TPR} = \frac{TP}{TP + FN}$$

And TNR measures the proportion of negative instances correctly identified:

$$\text{TNR} = \frac{TN}{TN + FP}$$

- Fairness Metric: To assess fairness, Equalized Odds was utilized, which quantifies the prediction error disparity between protected and non-protected groups with lower values indicating smaller disparities and thus higher fairness (Iosifidis & Ntoutsi, 2019). A more detailed explanation of Equalized Odds can be found in the related work section.

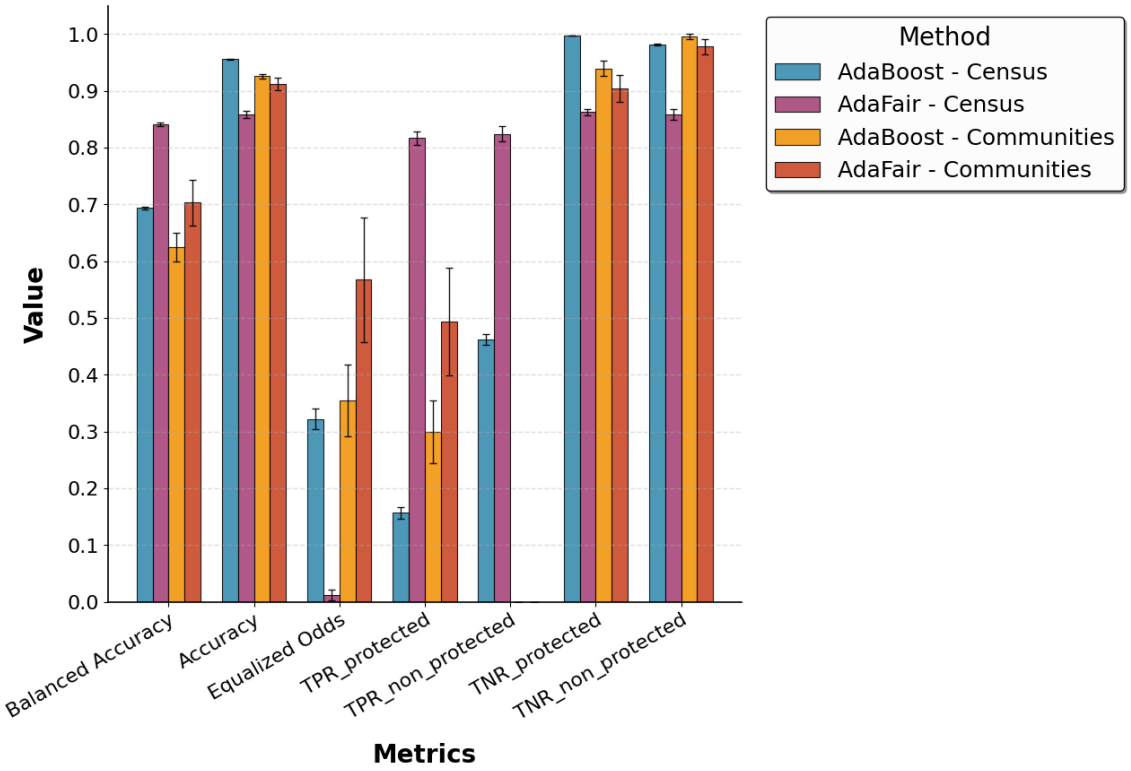
Additionally, the individual TPR_protected, TPR_non-protected, TNR_protected, and TNR_non-protected values were reported to provide a detailed breakdown of the model's performance on each subgroup. To further explore these metrics, box plots were made using the seaborn library, each showing the performance of one of the models (AdaBoost, AdaFair) in combination with one of the datasets. For example, the first plot shows the performance of AdaBoost on both data sets compared to the performance of AdaFair on these. With the X-axis indicating the different evaluation metrics and the y-axis the corresponding performance.

4 Results

The performance comparison between the AdaFair algorithm and the AdaBoost baseline on two datasets, KDD census income and Communities and Crime, is presented in Figure 1. As in the original AdaFair study (Iosifidis & Ntoutsi, 2019), predictive performance is evaluated using accuracy

and balanced accuracy, while fairness is assessed through Equalized Odds, as well as the true positive rate (TPR) and true negative rate (TNR) for both protected and non-protected groups.

Figure 1: Performance comparison of AdaFair and AdaBoost on KDD census income and Communities and Crime Datasets (Higher values reflect stronger predictive accuracy and fairness, whereas lower values are preferable for Equalized Odds).



On the KDD census income dataset, AdaFair outperforms AdaBoost in terms of balanced accuracy and fairness metrics, consistent with the findings of Iosifidis and Ntoutsi (2019). However, on the Communities and Crime dataset, while balanced accuracy remains aligned with expectations, AdaFair's fairness metrics deteriorate significantly compared to AdaBoost. This disparity is largely attributable to AdaFair's failure to correctly classify any positive instance of the non-protected group. As shown in Table 1, only 7 such instances exist in the Communities and Crime dataset, suggesting that the extreme proportion imbalance may have hindered AdaFair's ability to learn an effective classification for this subgroup. Notably, the TPR for the protected group also drops by 32 percentage points on the Communities and Crime dataset relative to the KDD census income dataset, indicating that the smaller dataset size may affect the algorithm's overall ability to generalize, particularly for minority classes.

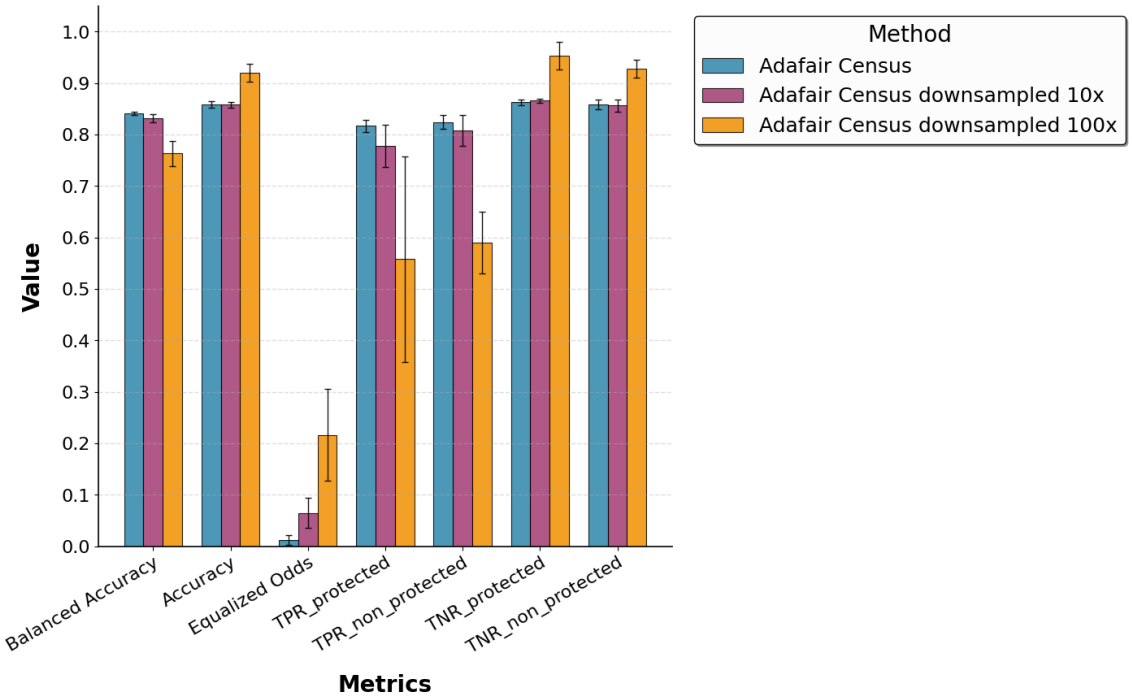
Table 1: Class Distribution by Dataset and Group

Dataset	Group	Positive	Negative
Communities and Crime	Protected	115	923
	Non-Protected	7	949
KDD census income	Protected	1,709	70,416
	Non-Protected	6,405	59,805

Figure 2 explores the impact of dataset size on the performance of AdaFair by evaluating the algorithm on the full KDD census income

dataset and downsampled versions by factors of 10 and 100. The results indicate that for the 10× downsampled dataset, neither predictive performance nor fairness metrics suffer a significant decline. However, in the 100× downsampled version, the Equalized Odds metric increases from 0,013 to 0,216, suggesting a significant drop in fairness. In this version, TNR values for both protected and non-protected groups increase, signaling potential overfitting to the majority class. These observations suggest the existence of a dataset size threshold below which AdaFair is unable to maintain fairness. Importantly, unlike in the Communities and Crime dataset, the TPR for the non-protected group in the 100× downsampled KDD census income dataset does not exhibit a significant decline, implying that the performance decrease observed in the Communities and Crime dataset is not solely due to proportion imbalance but likely influenced by other imbalances.

Figure 2: Impact of downsampling on AdaFair performance across KDD census income dataset sizes (Higher values reflect stronger predictive accuracy and fairness, whereas lower values are preferable for Equalized Odds).



To investigate the role of proportion imbalance further, upsampling was applied to the Communities and Crime dataset to artificially increase the number of positive instances of the non-protected group. The performance results of several upsampling strategies are summarized in Figures 3 and 4. Figure 3 illustrates the performance of different upsampling strategies: instance-based upsampling of the entire dataset, and targeted upsampling of only the positive non-protected group using either instance copying or the SMOTE algorithm. Figure 4 illustrates the performance across various upsampling rates of the instance copying method. The plot showing all upsampling versions is provided in Appendix A. Despite applying multiple upsampling techniques, including those specifically targeting the positive non-protected class, no meaningful improvement was observed in the TPR of the non-protected group or overall fairness. While slight performance gains were achieved through targeted upsampling of the positive non-protected group, these were insufficient to produce a substantive change in Equalized Odds and are likely attributable to random variation. Thus, the hypothesis that the proportion imbalance alone accounts for the poor performance of the positive non-protected class is not supported by the results.

Figure 3: AdaFair performance across different upsampling strategies for Communities and Crime dataset (Higher values reflect stronger predictive accuracy and fairness, whereas lower values are preferable for Equalized Odds).

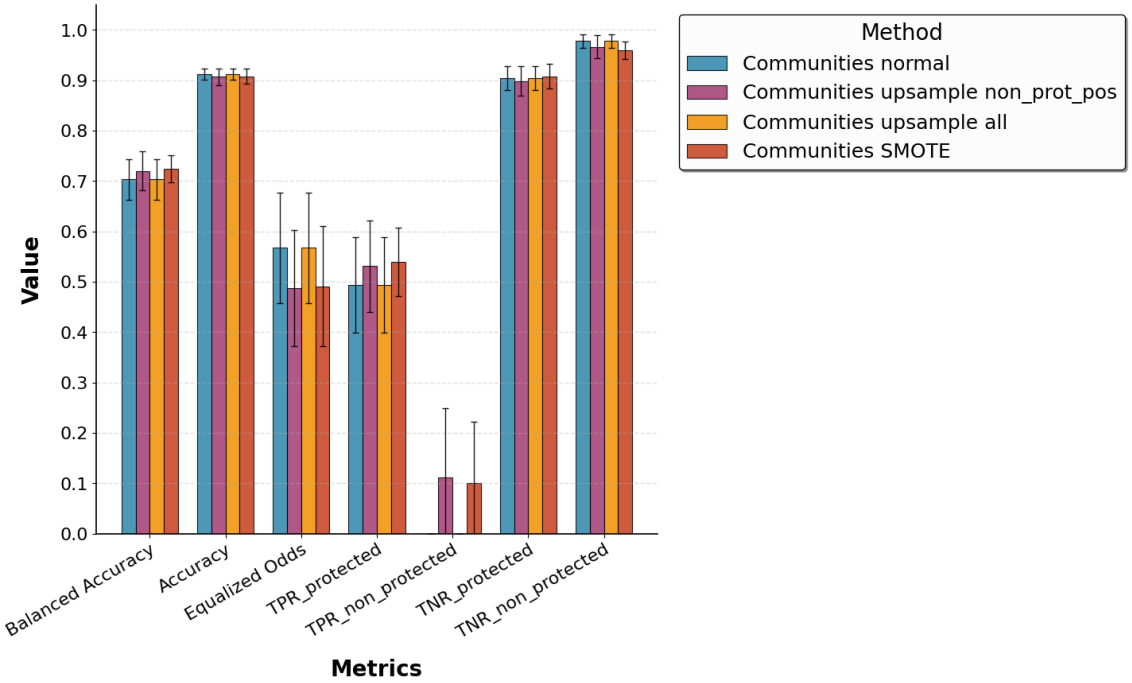
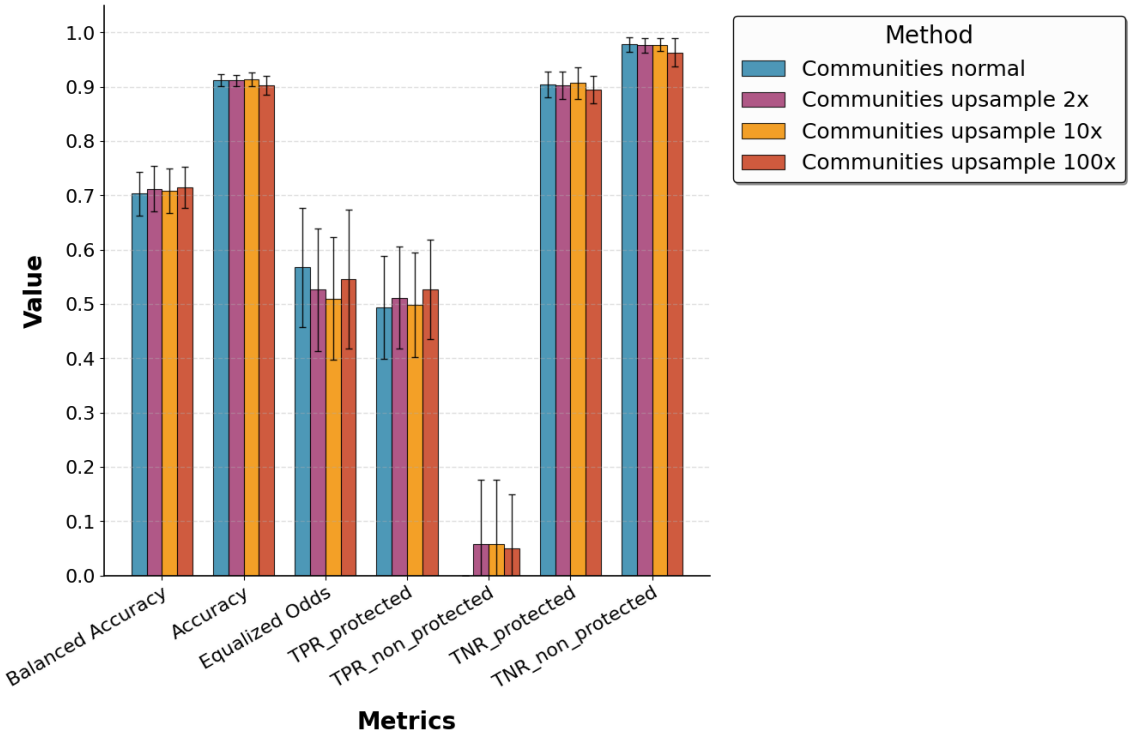


Figure 4: AdaFair performance across upsampling rates using instance copying for Communities and Crime dataset (Higher values reflect stronger predictive accuracy and fairness, whereas lower values are preferable for Equalized Odds).

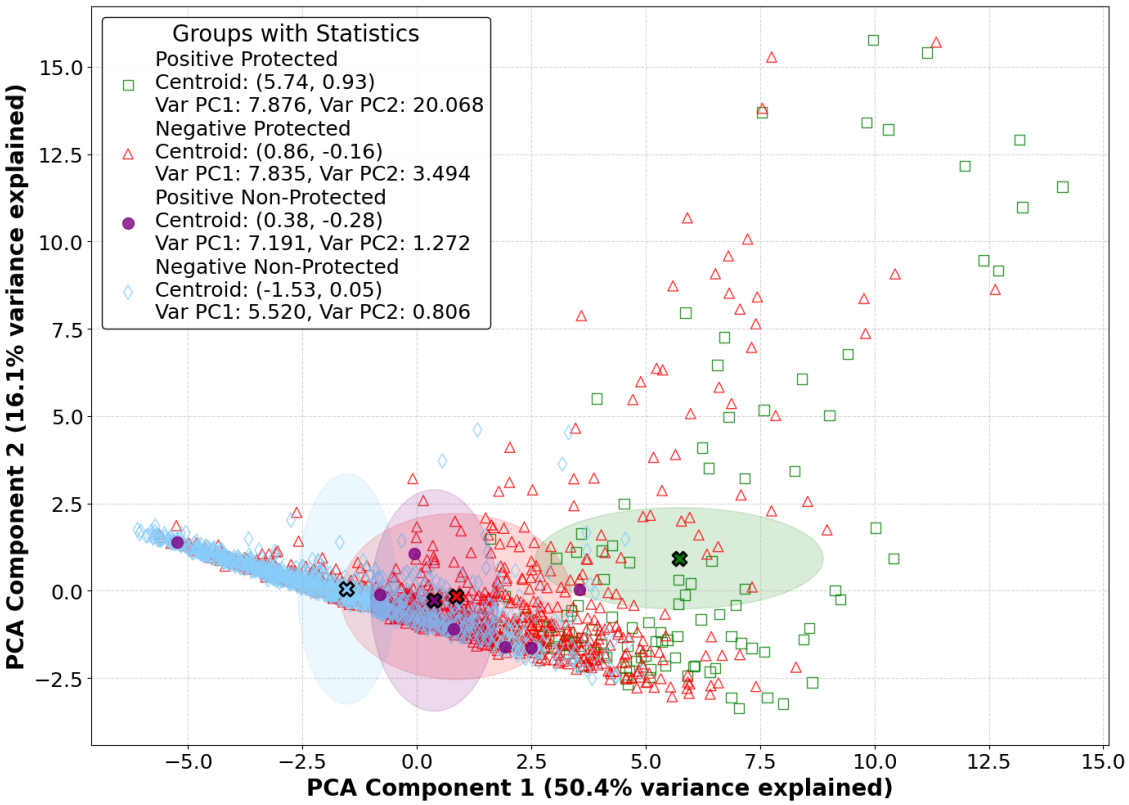


Figures 3 and 4 collectively indicate that the classification issue affecting the positive non-protected group in the Communities and Crime dataset is not resolved through simple rebalancing of the dataset. To further analyze this issue, PCA was performed on the Communities and Crime dataset, and the results are presented in Figure 5. The PCA projection displays the first two principal components for all subgroups, with centroids and variance values annotated. Each group is also represented by

a 95% confidence ellipse, based on the covariance of its projected data, illustrating the region within which the group's points are likely to fall. The positive non-protected group exhibits variances of 7.19 and 1.27 on the first and second principal components, respectively. These values are not substantially different from those observed in other groups: the positive protected group (7.88, 20.07), the negative protected group (7.83, 3.49), and the negative non-protected group (5.52,0.81). Therefore, variance imbalance does not appear to be a contributing factor. However, the spatial proximity between the centroid of the positive non-protected group and that of the negative protected group suggests a distance imbalance that may be more impactful. This overlap in feature space, combined with the previously identified proportion imbalance (7 vs. 923 samples), likely hampers AdaFair's ability to distinguish the positive non-protected group.

In summary, while AdaFair shows strong performance on the KDD census income dataset, its performance degrades on the Communities and Crime dataset, especially for the positive non-protected group. This degradation cannot be explained solely by class imbalance in terms of proportions. Rather, a more complex interaction between sample size, data distribution, and class separability, specifically, distance imbalance, appears to underlie the performance issues. These findings suggest that AdaFair may require enhanced strategies for handling nuanced distributional patterns for underrepresented subgroups.

Figure 5: Principal Component Analysis of Communities and Crime dataset by subgroup.



5 Discussion

The primary goal of this study was to evaluate the transferability of the AdaFair algorithm, a fairness-aware boosting algorithm, to small datasets, specifically focusing on its performance in terms of balanced accuracy and fairness (measured by Equalized Odds) under conditions of class imbalance. The investigation compared AdaFair's performance on the large KDD census income dataset (138,335 instances) and the smaller Communities and Crime dataset (1,994 instances), both characterized by severe class imbalance. Additionally, the study explored the impact of dataset size through downsampling the KDD census income dataset and upsampling the Communities and Crime dataset, aiming to understand how dataset size and class distribution affect AdaFair's ability to maintain fairness and predictive accuracy. The findings revealed critical insights into AdaFair's strengths and limitations, particularly its sensitivity to dataset size and complex distributional imbalances, which warrant a detailed discussion of the results, their implications, and potential improvements.

5.1 Evaluation of Findings

The results, as presented in Figure 1, confirmed AdaFair's superior performance on the KDD census income dataset, aligning with the original findings (Iosifidis & Ntoutsi, 2019). AdaFair outperformed the standard AdaBoost algorithm in both balanced accuracy and fairness metrics, demonstrating its ability to effectively balance predictive performance with fairness in a large, imbalanced dataset. However, the performance on the Communities and Crime dataset was unexpectedly poor, particularly in terms of fairness. AdaFair failed to correctly classify any positive instances in the non-protected group, which consisted of only seven instances, leading to a significant deterioration in Equalized Odds compared to AdaBoost. This finding was surprising, as AdaFair was designed to handle class imbalance, yet it struggled with the extreme scarcity of positive non-protected instances in the Communities and Crime dataset.

The drop in fairness metrics on the Communities and Crime dataset, coupled with a 32 percentage point decrease in the TPR for the protected group compared to the KDD census income dataset, suggests that dataset size and class distribution play a critical role in AdaFair's performance. The KDD census income dataset's large size likely provided sufficient samples across all subgroups, enabling AdaFair to learn robust decision boundaries. In contrast, the Communities and Crime dataset's small size and extreme imbalance, particularly in the positive non-protected group, may have limited the algorithm's ability to generalize, leading to overfitting to the majority class or failure to learn effective patterns for minority subgroups.

To further explore the role of dataset size, Figure 2 examined AdaFair's performance on downsampled versions of the KDD census income dataset (10× and 100× reductions). The 10× downsampled dataset showed minimal degradation in predictive performance and fairness, indicating that AdaFair can tolerate moderate reductions in dataset size. However, the 100× downsampled dataset exhibited a noticeable increase in Equalized Odds, signaling a sharp decline in fairness. This suggests a critical threshold below which AdaFair's fairness-aware mechanism becomes ineffective, likely due to insufficient data to reliably classify the minority class. Interestingly, the TPR for the non-protected group in the 100×

downsampled KDD census income dataset did not decline as severely as in the Communities and Crime dataset, implying that this dataset's poor performance was not solely due to proportion imbalance but rather a combination of factors, including dataset size and other distributional imbalances.

Figures 3 and 4 investigated whether upsampling the Communities and Crime dataset could mitigate the observed fairness issues. Various strategies, including upsampling the entire dataset, targeting the positive non-protected group, and using SMOTE, failed to significantly improve Equalized Odds and the TPR for the non-protected group. This was unexpected, as upsampling is a common preprocessing technique for addressing class imbalance (Hort et al., 2024). The lack of improvement suggests that proportion imbalance alone does not fully explain AdaFair's struggles with the Communities and Crime dataset. To probe deeper, the PCA in Figure 5 revealed that variance imbalance was not a significant factor, as the variance of the positive non-protected group was comparable to that of the other groups. However, a notable distance imbalance was observed, with the positive non-protected group's centroid overlapping with the negative protected group's centroid in feature space. This overlap, combined with the extreme proportion imbalance between these two groups (7 vs. 923 samples), likely confused AdaFair's decision boundaries, hindering its ability to distinguish the positive non-protected group.

5.2 Limitations and Possible Solutions

While AdaFair performed well on the large KDD census income dataset, its failure on the smaller Communities and Crime dataset highlights significant limitations in handling extreme class imbalances and small dataset sizes. One potential weakness lies in the choice of categorical encoding for preprocessing. The study used label encoding for categorical features, which may introduce ordinal assumptions and affect model performance. An alternative, such as one-hot encoding, could preserve categorical distinctions but risks increasing dimensionality, potentially exacerbating issues in small datasets. Future work could explore the effects of the encoding techniques on AdaFair's performance.

Another limitation is the study's focus on proportion imbalance, as defined by Wu's taxonomy (2023), without fully exploring other imbalance types (e.g., variance, distance, neighborhood, quality). The PCA results suggest that distance imbalance significantly contributed to AdaFair's poor performance on the Communities and Crime dataset. Future studies could incorporate preprocessing techniques to address this distance imbalance and see if this improves the performance on the positive non-protected group. Additionally, the study's reliance on Equalized Odds as the sole fairness metric may have limited its ability to capture other dimensions of fairness, such as Demographic Parity or individual fairness. Given that the literature identifies over 100 fairness metrics (Hort et al., 2024), exploring alternative measures, such as Statistical Parity or Equal Opportunity, both of which have corresponding AdaFair variants (Iosifidis & Roy, 2023), could offer a more comprehensive evaluation of fairness trade-offs.

The upsampling strategies employed were also limited by their simplicity. While SMOTE was used, it may not have been effective in generating meaningful synthetic samples for the positive non-protected group due to its small size and overlap with other groups. Advanced generative methods, such as variational autoencoders or generative adversarial networks, could produce more diverse and representative synthetic samples, potentially improving AdaFair's performance.

Moreover, the use of postprocessing techniques in combination with AdaFair, which might address the fairness dropoff, was not investigated. Finally, the study did not explore hyperparameter tuning for AdaFair, such as adjusting the number of estimators or fairness weight parameters, which could enhance its robustness to small datasets.

5.3 Contextualization and Contribution

The findings contribute to the growing body of research on fairness-aware machine learning by highlighting the challenges of transferring fairness-aware algorithms across datasets with differing sizes and distributional properties. The study aligns with prior work, noting the difficulty of applying fairness-aware algorithms to small datasets, and extends this by demonstrating that AdaFair's performance is sensitive not only to proportion imbalance but also to distance imbalance. This nuanced understanding of imbalance types, as per Wu's taxonomy (2023), underscores the need for more robust fairness-aware algorithms that can handle complex distributional patterns.

The contribution of this study lies in its empirical demonstration of AdaFair's limitations in small, imbalanced datasets and its identification of distance imbalance as a critical factor affecting fairness. By comparing AdaFair's performance across datasets and exploring upsampling and downsampling, the study provides insights for practitioners seeking to deploy fairness-aware algorithms in data-scarce environments. It also highlights the importance of considering multiple types of imbalances beyond proportion, offering a more comprehensive framework for evaluating fairness-aware algorithms.

5.4 Future Research Directions

Future research should focus on enhancing AdaFair's robustness to small datasets and complex imbalances. Exploring alternative fairness metrics, such as Statistical Parity or Equal Opportunity, could provide a more holistic view of AdaFair's performance and show the ability to handle certain types of imbalances better. Additionally, testing AdaFair on datasets with varying levels of variance, distance, or quality imbalances could further clarify its limitations. Moreover, postprocessing techniques could be utilized to shift the decision boundary, potentially mitigating distance imbalance. Finally, integrating advanced generative techniques for data augmentation and conducting extensive hyperparameter optimization could enhance AdaFair's applicability in data-scarce environments.

6 Conclusion

This study aimed to assess the transferability of the AdaFair algorithm to small datasets, focusing on its ability to maintain balanced accuracy and fairness (via Equalized Odds) in the presence of severe class imbalance. The results showed that AdaFair excels on the large KDD census income dataset, outperforming Adaboost in both predictive performance and fairness, consistent with prior work (Iosifidis & Ntoutsi, 2019). However, its performance degraded significantly on the smaller Communities and Crime dataset, particularly for the positive non-protected group, due to a combination of extreme proportion imbalance and distance imbalance. Upsampling strategies failed to mitigate these issues, and downsampling the KDD census income dataset revealed a

dataset size threshold below which fairness deteriorates.

These findings contribute to the fair machine learning literature by highlighting the challenges of applying fairness-aware algorithms in data-scarce environments and identifying distance imbalance as a key factor affecting performance. The implications for the field include the need for more sophisticated preprocessing techniques and fairness-aware algorithms that can handle multiple types of imbalances. Future research should therefore explore advanced data augmentation, alternative fairness metrics, and hyperparameter tuning to enhance AdaFair's robustness.

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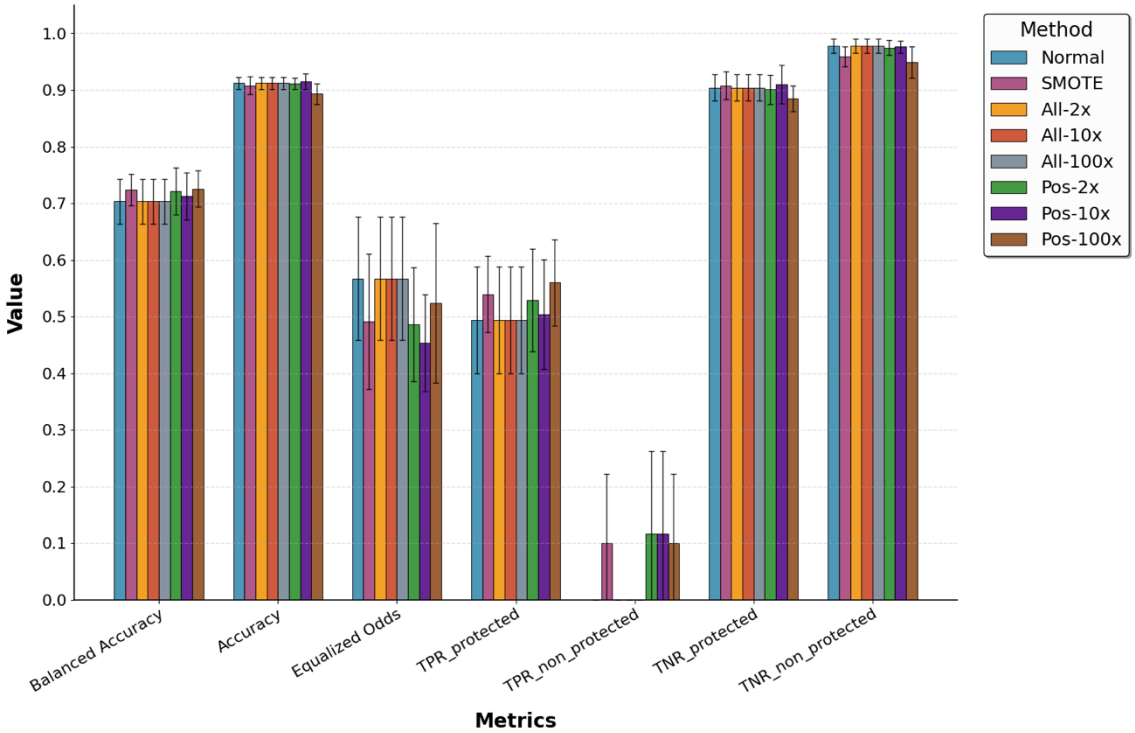
Appendix A: Comprehensive performance of AdaFair across all upsampling versions for Communities and Crime dataset (Higher values indicate better predictive and fairness performance, while lower values are better for Equalized Odds).